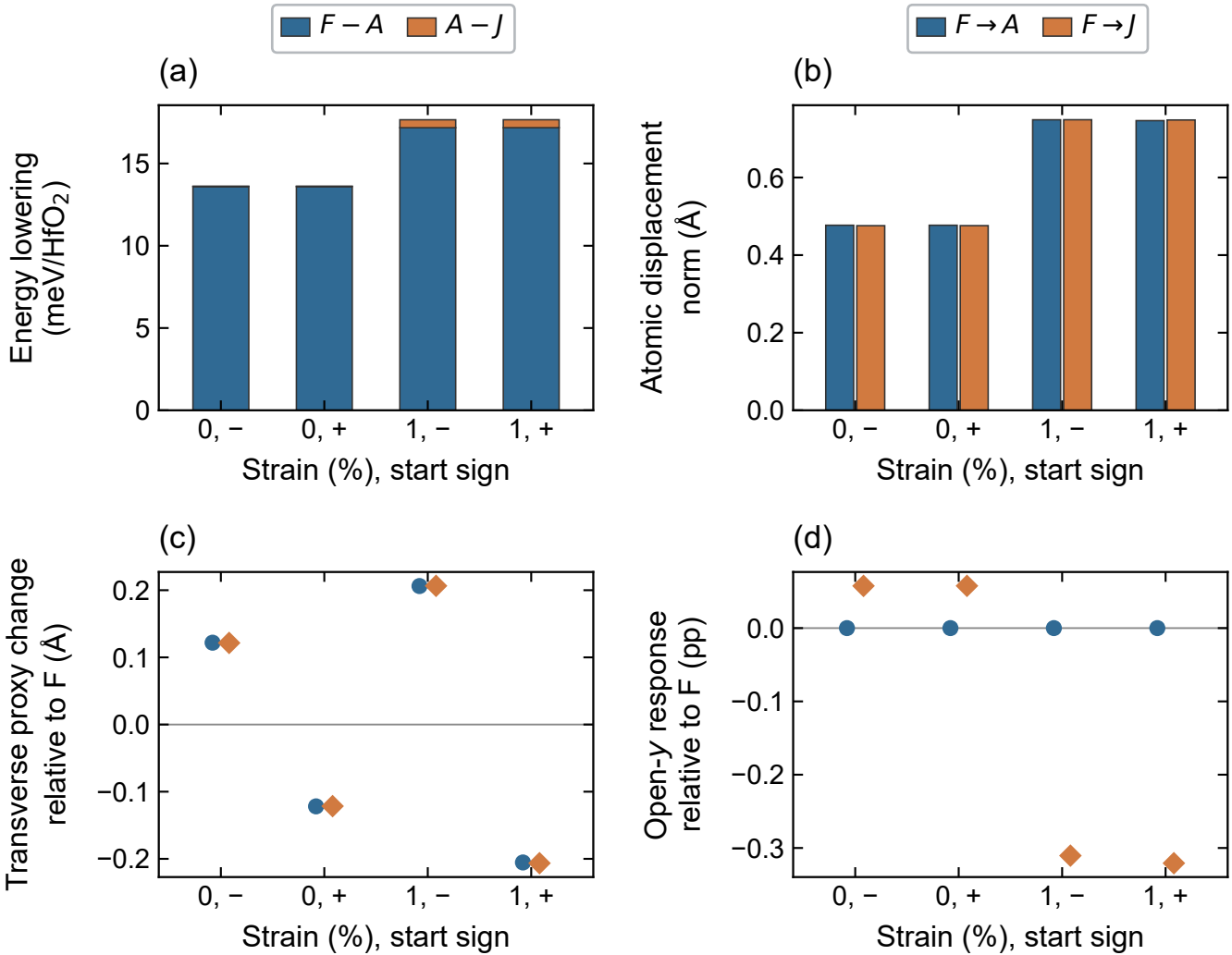
F − A
A − J
(a)
Energy lowering (meV/HfO2)
0
5
10
15
0, −
0, +
1, −
1, +
Strain (%), start sign
F → A
F → J
(b)
Atomic displacement norm (Å)
0.0
0.2
0.4
0.6
0, −
0, +
1, −
1, +
Strain (%), start sign
(c)
Transverse proxy change relative to F (Å)
0.2
0.1
0.0
−0.1
−0.2
0, −
0, +
1, −
1, +
Strain (%), start sign
(d)
Open-y response relative to F (pp)
0.0
−0.1
−0.2
−0.3
0, −
0, +
1, −
1, +
Strain (%), start sign